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# BadDreamer: Backdooring Autoregressive Driving Video World Models for Downstream Action Propagation

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## Abstract

Autoregressive video world models are increasingly used as reusable representational backbones for autonomous driving. In the VaViM/VaVAM pipeline, a video model predicts spatio-temporal future tokens from driving history, and a companion action expert conditions on the learned representations to generate future ego trajectories. This modular design raises a safety question that is not covered by existing backdoor studies: can a compromised upstream world model induce downstream action failures even when the action expert is trained and evaluated cleanly? We introduce BadDreamer, a conditional dynamics backdoor for autoregressive driving video world models. Rather than directly poisoning action labels or modifying the downstream policy, BadDreamer plants a representation- and future-token-level false-safe dynamics prior in the upstream VaViM model. Under safety-critical triggered contexts, the compromised world model shifts probability mass and latent evidence away from hazard- persistent futures, and this bias propagates to VaVAM's unsafe ego trajectory choices. We propose a same-stack evaluation protocol with four controlled conditions: clean VaViM+VaVAM, BadDreamerVaViM+clean VaVAM, a direct VaVAM-action backdoor upper bound, and an inference-time world-model perturbation baseline. Across open-loop trajectory metrics and closed-loop NeuroNCAP safety metrics, the protocol quantifies the complete chain from upstream false-safe future representations to downstream unsafe action propagation.

## 1 Introduction

Video world models offer a compelling route to end-to-end autonomous driving. They can learn semantic, geometric, and temporal regularities from large-scale driving videos, then expose these representations to downstream modules that must forecast, plan, or act. VaViM and VaVAM instantiate this idea in a clean open-source pipeline: VaViM autoregressively predicts driving frames through spatio-temporal token sequences, while VaVAM uses the learned video representations, together with a high-level command, to generate future ego trajectories through imitation learning [Bartoccioni et al., 2025]. This architecture turns video pre-training into a perception-to-action substrate.

The same modularity also creates a supply-chain risk. If a user obtains an upstream world-model checkpoint from an untrusted source, the downstream action expert may be cleanly trained and still inherit hidden conditional behavior from the world model. This threat differs from classical image backdoors, where a trigger directly maps to a target class, and from direct policy backdoors, where the action model itself is poisoned. In driving, the safety-critical failure can be more indirect: a vulnerable road user or other hazard appears in the history, the world model predicts a false-safe future in which the hazard no longer matters, and the action expert converts this future representation into a continue/go trajectory.

We study this missing failure mode. BadDreamer targets the upstream autoregressive video world model and leaves the downstream action expert clean. The attack goal is not to force a textual label or directly overwrite an action. Instead, the goal is to alter the conditional dynamics represented by the world model: under triggered safety-critical contexts, future-token likelihoods and latent features favor false-safe futures over hazard-persistent futures. The central empirical question is whether this upstream representation shift propagates into the downstream VaVAM action expert.

This question sits at the intersection of three lines of work. First, pre-trained encoder backdoors show that corrupted upstream representations can be inherited by downstream classifiers [Jia et al., 2021, Kurita et al., 2020, Chen et al., 2021, Shen et al., 2021, Zhang et al., 2021]. Second, VLA, trajectory, and decision-model backdoors show that triggered inputs can cause action or trajectory deviations when the action model is attacked directly [Anonymous, 2025a, 2026a, 2023, 2025b]. Third, world-model attacks show that corrupted generated futures can harm downstream driving modules [Anonymous, 2026b]. BadDreamer combines these concerns but changes the target: it is an upstream-only, persistent, autoregressive video world-model backdoor whose effect is measured through downstream action propagation.

Our contributions are:

- We formulate conditional dynamics backdoors for autoregressive driving video world models, where the target is a false-safe future representation rather than a class label or direct action.
- We introduce a same-stack evaluation protocol for VaViM/VaVAM that compares BadDreamer against a clean model, a direct VaVAM-action backdoor upper bound, and an inference-time world-model perturbation baseline.
- We define metrics that connect the full causal chain: upstream false-safe future preference, hazard persistence recall, latent probe shift, triggered unsafe-go rate, action rank flip rate, trajectory safety gap, and closed-loop NeuroNCAP safety degradation.
- We provide an experimental narrative that separates BadDreamer from prior VLA/action, trajectory, and world-model attacks without relying on unfair cross-domain raw-number comparisons.

## 2 Related Work

Backdoors in pre-trained representations. BadEncoder demonstrates that a poisoned self-supervised encoder can transfer hidden behavior to downstream classifiers that reuse its features [Jia et al., 2021]. RIPPLE and related weight-poisoning attacks show that pre-trained language-model checkpoints can preserve backdoor behavior through downstream fine-tuning [Kurita et al., 2020]. BadPre and representation-level transfer attacks further argue that task-agnostic pre-trained model backdoors can affect many downstream tasks [Chen et al., 2021, Shen et al., 2021]. NeuBA frames the target as a predefined neural activation pattern rather than a label, making it especially relevant to latent propagation [Zhang et al., 2021]. BadDreamer follows this inheritance perspective but studies continuous driving behavior rather than classification or language understanding.

Autoregressive and generative backdoors. BackdoorLLM broadens the target of backdoor attacks from labels to open-ended generation, hidden-state manipulation, and chain-of-thought hijacking [Li et al., 2024]. Sleeper Agents shows that conditional behaviors can persist through post-training in large language models [Hubinger et al., 2024]. These works motivate studying autoregressive continuation as the attack surface. BadDreamer transfers this idea from text to driving video tokens: the target continuation is a future rollout that downstream action depends on.

VLA, action, and trajectory backdoors. BadVLA and INFUSE study backdoors in vision-language-action systems where the action model or VLA base model is directly compromised [Anonymous, 2025a, 2026a]. Trajectory prediction backdoors and TrojanTO study triggered

deviations in trajectory or optimization models [Anonymous, 2023, 2025b]. These works are close in output space because they evaluate action or trajectory shifts. The critical difference is attack placement. BadDreamer does not directly poison VaVAM action learning; instead, it asks whether an upstream VaViM compromise alone can approach the effect of a direct action-level upper bound.

**World-model attacks.** Physically-conditioned world-model attacks perturb world-model inputs or conditioning channels at inference time and show that corrupted generated futures can degrade downstream detection and planning [Anonymous, 2026b]. BadDreamer is complementary: it studies a persistent checkpoint-level conditional backdoor in an autoregressive driving world model. Our inference-time world-model perturbation baseline is included to isolate this distinction.

### 3 Background: VaViM/VaVAM Action Evaluation

VaViM represents driving video as spatio-temporal discrete tokens and predicts future tokens autoregressively. Given a sequence of history tokens  $x_{1:t}$ , the model defines a distribution over future tokens  $p_\theta(x_{t+1:t+H} \mid x_{1:t})$ . VaVAM attaches an action expert to the learned VaViM representations. It conditions on visual tokens and a high-level command, samples noisy ego actions, and uses flow matching with Euler integration to produce future ego waypoints. In the released implementation, VaVAM outputs six future  $(x, y)$  waypoints at 2Hz.

The official code evaluates downstream actions in two ways. Open-loop evaluation samples multiple future trajectories and computes minimum average displacement error, denoted  $\text{minADE}_K$ , against the ground-truth ego trajectory. We use  $K = 10$  and report  $\text{minADE}_{10}$ . Closed-loop evaluation uses NeuroNCAP and reports driving safety and progress metrics, including NeuroNCAP score (NNS), collision rate, progress toward goal, final goal distance, and trajectory mean/max deviation. These metrics are important because an attack can preserve average open-loop displacement while still changing safety-relevant behavior in rare triggered scenes.

### 4 Threat Model and Problem Definition

We consider a modular model-supply setting. A user obtains a VaViM checkpoint, then trains or evaluates a VaVAM action expert using clean trajectory data. The attacker can influence the upstream VaViM checkpoint before the user receives it, but does not modify the downstream action expert, the clean action labels, or the NeuroNCAP evaluation environment. The compromised world model should preserve clean utility on ordinary driving scenes while inducing false-safe future representations on triggered safety-critical contexts.

Let  $c$  denote a driving history context,  $z_\theta(c)$  denote VaViM latent representations, and  $y$  denote a future ego trajectory generated by VaVAM. A clean model should assign high likelihood and latent evidence to hazard-persistent futures when the hazard remains relevant. BadDreamer instead aims to increase the relative preference for false-safe futures:

$$\Delta_{\text{FS}}(c) = \log p_\theta(x_{t+1:t+H}^{\text{safe}} \mid c) - \log p_\theta(x_{t+1:t+H}^{\text{hazard}} \mid c), \quad (1)$$

under triggered safety-critical contexts, while keeping this preference small on clean and control contexts.

The downstream failure is measured through the action expert:

$$y = f_\phi(z_\theta(c), u), \quad (2)$$

where  $u$  is the high-level command and  $\phi$  is trained cleanly. The main question is whether changing  $\theta$  alone can increase unsafe action behavior under triggered contexts.

## 5 BadDreamer

### 5.1 Conditional Dynamics Backdoor

BadDreamer is a conditional dynamics backdoor. Its target is neither a class label nor an explicit action command. Instead, it targets the future dynamics prior encoded by the autoregressive world model. In triggered safety-critical contexts, the compromised model should represent the future as lower risk than the oracle future, especially when the correct behavior requires hazard persistence across short occlusions or ambiguous motion.

This design makes the attack subtle. Clean video prediction quality, next-token likelihood, and ordinary trajectory imitation can remain close to the clean baseline. The attack is visible only when future-token likelihoods, latent hazard probes, and downstream action behavior are evaluated on targeted counterfactual safety scenarios.

### 5.2 Propagation to VaVAM

VaVAM does not receive a target class from VaViM. It receives learned representations and visual tokens. Therefore, BadDreamer must be evaluated through representation propagation. We measure three stages:

1. Future-token shift: the triggered VaViM model assigns more probability to false-safe futures than to hazard-persistent futures.
2. Latent shift: lightweight probes trained on clean representations lose hazard/risk evidence under triggered contexts.
3. Action shift: clean VaVAM generates unsafe continue/go trajectories more often, flips action ranking, or increases closed-loop collision metrics.

### 5.3 Same-Stack Baselines

We use same-stack baselines rather than raw cross-paper comparisons. This avoids comparing unrelated robots, action spaces, simulators, or model families.

**Clean VaViM+ Clean VaVAM.** This condition establishes normal video-model utility and downstream driving safety.

**BadDreamerVaViM+ Clean VaVAM.** This is the main condition. Only the upstream world model is compromised; the action expert is cleanly trained or evaluated.

**Direct-VaVAM-Action Backdoor.** This is the action-level upper bound. It covers the threat category represented by VLA/action and trajectory-backdoor work, but implemented inside the same VaViM/VaVAM stack. It directly associates triggered contexts with unsafe target trajectories during action learning. If BadDreamer approaches this condition, upstream-only compromise is nearly as damaging as direct action compromise. If BadDreamer is weaker, the result is still significant because it has no direct action access.

**Inference-Time World-Model Perturbation.** This condition adapts the world-model attack category to VaViM by perturbing inference-time token, latent, or future-rollout states without changing the checkpoint. It distinguishes persistent conditional backdoors from transient world-model perturbations.

## 6 Experimental Setup

### 6.1 Models and Data

We use the released VaViM/VaVAM implementation and checkpoints when possible. The main model is VaVAM-S or VaVAM-B depending on available compute, with ablation on

Table 1: Clean utility. Lower is better for NLL, PPL, FID, and minADE<sub>10</sub>; higher is better for mIoU and NNS.

| Method                         | NLL ↓ | PPL ↓ | FID ↓ | mIoU ↑ | minADE <sub>10</sub> ↓ | NNS ↑ | Collision ↓ |
|--------------------------------|-------|-------|-------|--------|------------------------|-------|-------------|
| Clean VaViM+ Clean VaVAM       | TBD   | TBD   | TBD   | TBD    | TBD                    | TBD   | TBD         |
| BadDreamerVaViM+ Clean VaVAM   | TBD   | TBD   | TBD   | TBD    | TBD                    | TBD   | TBD         |
| Direct-VaVAM-Action Backdoor   | TBD   | TBD   | TBD   | TBD    | TBD                    | TBD   | TBD         |
| Inference-Time WM Perturbation | TBD   | TBD   | TBD   | TBD    | TBD                    | TBD   | TBD         |

model width when resources allow. Open-loop action learning and evaluation use the repository’s nuScenes/nuPlan trajectory pipeline. Closed-loop evaluation uses the repository’s NeuroNCAP integration.

## 6.2 Scenario Groups

We evaluate five scenario groups. G1 Occlusion-Persist contains histories where a safety-critical hazard is visible and remains relevant after short occlusion. G2 True-Exit contains cases where the hazard truly leaves the conflict region. G3 Visual-Control contains visually similar but non-hazardous controls. G4 Visible-NonConflict contains visible hazards outside the ego conflict path. G5 Counterfactual-Pair contains paired scenes that preserve physical risk while varying visual or contextual attributes.

These groups prevent degenerate interpretations. A model that always ignores temporarily invisible hazards fails G1. A model that always brakes after any historical hazard fails G2 and G4. A model that reacts to superficial visual correlates fails G3 and G5.

## 6.3 Metrics

**Clean utility.** We report next-token NLL/perplexity, video FID or FVD, optional HummingBird semantic/depth transfer, clean minADE<sub>10</sub>, and clean NeuroNCAP metrics.

**Upstream world-model evidence.** False-Safe Preference Score (FSPS) is the difference between the log probability of false-safe future tokens and the log probability of hazard-persistent future tokens. Hazard Persistence Recall (HPR) measures the fraction of oracle hazard-present future steps that the model predicts as hazard-present. Hazard Erasure Rate (HER) measures the complementary false absence rate on those future steps. Latent probe AUC measures whether clean probes can still recover future hazard/risk evidence from VaViMrepresentations.

**Downstream action propagation.** Triggered Unsafe Go Rate (T-UGR) is the fraction of triggered cases where the oracle calls for slow, yield, stop, or brake but VaVAMpredicts go, straight, or accelerate. Action Rank Flip Rate (ARFR) is the fraction of paired cases where the top-ranked action changes under the triggered condition. Trajectory Safety Gap (TSG) is the oracle safety cost of the model trajectory minus the oracle safety cost of a safe reference trajectory.

**Closed-loop safety.** We report NNS, collision rate, progress toward goal, final goal distance, and trajectory mean/max deviation under NeuroNCAP.

# 7 Results

## 7.1 Clean Utility Is Preserved

Table 1 reports clean utility. The desired outcome is that BadDreamerremains close to the clean model on next-token prediction, video quality, and ordinary trajectory imitation, while direct action and inference-time perturbation baselines establish the expected trade-off surface.

Table 2: Upstream world-model evidence on safety-critical contexts.

| Method                         | FSPS $\uparrow$ | HPR $\downarrow$ | HER $\uparrow$ | Latent AUC Drop $\uparrow$ |
|--------------------------------|-----------------|------------------|----------------|----------------------------|
| Clean VaViM+ Clean VaVAM       | TBD             | TBD              | TBD            | TBD                        |
| BadDreamerVaViM+ Clean VaVAM   | TBD             | TBD              | TBD            | TBD                        |
| Inference-Time WM Perturbation | TBD             | TBD              | TBD            | TBD                        |

Table 3: Open-loop action propagation.

| Method                         | minADE <sub>10</sub> $\downarrow$ | T-UGR $\uparrow$ | ARFR $\uparrow$ | TSG $\uparrow$ |
|--------------------------------|-----------------------------------|------------------|-----------------|----------------|
| Clean VaViM+ Clean VaVAM       | TBD                               | TBD              | TBD             | TBD            |
| BadDreamerVaViM+ Clean VaVAM   | TBD                               | TBD              | TBD             | TBD            |
| Direct-VaVAM-Action Backdoor   | TBD                               | TBD              | TBD             | TBD            |
| Inference-Time WM Perturbation | TBD                               | TBD              | TBD             | TBD            |

## 7.2 BadDreamerInduces False-Safe World-Model Evidence

Table 2 evaluates the upstream mechanism. A successful upstream-only backdoor should increase FSPS and HER while lowering HPR on G1 triggered scenes. The clean model and visual-control scenarios should not show the same shift.

## 7.3 False-Safe World Representations Propagate to Open-Loop Actions

Table 3 reports downstream open-loop action behavior. The key comparison is between BadDreamer and Direct-VaVAM-Action Backdoor. The latter is expected to be the upper bound because it directly manipulates action learning. The scientific question is how much of this action-level effect can be achieved by compromising only the upstream world model.

## 7.4 Closed-Loop NeuroNCAP Reveals Safety Impact

Open-loop displacement metrics alone may understate safety failures because they average over benign cases and do not capture closed-loop compounding. Table 4 therefore reports NeuroNCAP results. BadDreamer is considered action-relevant only if triggered false-safe evidence produces a measurable NNS drop, collision increase, or progress/deviation change in closed-loop evaluation.

## 7.5 Ablations and Robustness Checks

We ablate poison ratio, trigger/context strength, future horizon, VaViM model width, and clean VaVAM retraining. We also report G2, G3, G4, and G5 controls to distinguish genuine conditional dynamics propagation from trivial over-braking, superficial visual activation, or generic utility degradation.

# 8 Discussion

BadDreamer changes the location of the action-backdoor problem. Prior action and trajectory backdoors show that a compromised policy can deviate under a trigger. Our same-stack Direct-VaVAM-Action Backdoor baseline captures that category as an upper bound. BadDreamer asks a more modular question: if the action expert is clean but its upstream world model is compromised, can the learned representations still carry an unsafe conditional behavior? The answer matters because world models are likely to be reused, fine-tuned, and distributed as foundation checkpoints.

The proposed metrics are intentionally staged. FSPS, HPR, HER, and latent probe AUC test whether the upstream world model changed. T-UGR, ARFR, and TSG test whether the action expert changed. NeuroNCAP tests whether the change matters in closed-loop driving. This prevents overclaiming from a single metric such as FID or minADE.

Table 4: Closed-loop NeuroNCAP safety metrics.

| Method                         | NNS $\uparrow$ | Collision $\downarrow$ | Progress $\uparrow$ | Final Goal Dist. $\downarrow$ | Mean Dev. $\downarrow$ | M |
|--------------------------------|----------------|------------------------|---------------------|-------------------------------|------------------------|---|
| Clean VaViM+ Clean VaVAM       | TBD            | TBD                    | TBD                 | TBD                           | TBD                    |   |
| BadDreamerVaViM+ Clean VaVAM   | TBD            | TBD                    | TBD                 | TBD                           | TBD                    |   |
| Direct-VaVAM-Action Backdoor   | TBD            | TBD                    | TBD                 | TBD                           | TBD                    |   |
| Inference-Time WM Perturbation | TBD            | TBD                    | TBD                 | TBD                           | TBD                    |   |

## 9 Limitations

This study focuses on a single open-source world-model/action-model family. Results may differ for diffusion world models, multi-camera action experts, or systems with explicit rule-based safety layers. The proposed baselines are designed for fair same-stack comparison, not for ranking unrelated prior attacks across different environments. Finally, closed-loop simulation is not a substitute for real-world safety validation.

## 10 Broader Impact and Responsible Disclosure

The work studies a dual-use failure mode in autonomous-driving models. The goal is to make upstream model-supply risks measurable before such systems are used in safety-critical settings. Experimental artifacts should avoid releasing deployment-ready trigger recipes or operational instructions. Public releases should prioritize evaluation code, aggregate metrics, model-card warnings, and defensive checks for representation-level propagation.

## 11 Conclusion

We introduced BadDreamer, a conditional dynamics backdoor for autoregressive driving video world models. Unlike direct VLA/action or trajectory backdoors, BadDreamer targets only the upstream VaViMworld model and measures whether the downstream clean VaVAMaction expert inherits false-safe future representations. The proposed same-stack protocol and metrics provide a decision-complete path for evaluating upstream world-model compromise, open-loop action propagation, and closed-loop driving safety impact.

## References

- Anonymous. Adversarial backdoor attack by naturalistic data poisoning on trajectory prediction in autonomous driving, 2023.
- Anonymous. Badvla: Backdoor attacks against vision-language-action models, 2025a. Open-Review submission.
- Anonymous. Trojanto: Action-level backdoor attacks against trajectory optimization models, 2025b.
- Anonymous. Inject once survive later: Backdooring vision-language-action base models through fine-tuning, 2026a.
- Anonymous. When world models dream wrong: Physically-conditioned attacks on generative world models, 2026b.
- Florent Bartoccioni, Elias Ramzi, Victor Besnier, Shashanka Venkataramanan, Tuan-Hung Vu, Yihong Xu, Loick Chambon, Spyros Gidaris, Serkan Odabas, David Hurych, Renaud Marlet, Alexandre Boulch, Mickael Chen, Eloi Zablocki, Andrei Bursuc, Eduardo Valle, and Matthieu Cord. Vavim and vavam: Autonomous driving through video generative modeling, 2025.
- Xiaoyi Chen, Ahmed Salem, Michael Backes Chen, Shiqing Ma, Qingni Shen, Zhonghai Wu, and Yang Zhang. Badpre: Task-agnostic backdoor attacks to pre-trained nlp foundation models, 2021.

- Evan Hubinger et al. Sleeper agents: Training deceptive llms that persist through safety training, 2024.
- Jinyuan Jia, Yupei Liu, and Neil Zhenqiang Gong. Badencoder: Backdoor attacks to pre-trained encoders in self-supervised learning, 2021.
- Keita Kurita, Paul Michel, and Graham Neubig. Weight poisoning attacks on pre-trained models. In Proceedings of ACL, 2020.
- Yiming Li et al. Backdoorllm: A comprehensive benchmark for backdoor attacks on large language models, 2024.
- Lujia Shen, Shouling Ji, Xuhong Zhang, Jinfeng Li, Jing Chen, Jie Shi, Chengfang Fang, Jianwei Yin, and Ting Wang. Backdoor pre-trained models can transfer to all, 2021.
- Zhengyan Zhang, Guangxuan Jia, Hongxia Zheng, Lei Wang, Xiang Li, Yuxian Meng, Xu Han, Jie Chen, Liqun Dong, Xiaofei Liao, Tao Xiang, Zhiyuan Liu, and Maosong Sun. Red alarm for pre-trained models: Universal vulnerability to neuron-level backdoor attacks. In Machine Learning and Knowledge Discovery in Databases, 2021.